

2-Mark Questions & Answers -- Data Structures Module 2: Stacks & Queues

Q1. Define Stack ADT and list its primary operations.

A Stack is a linear data structure following the Last-In, First-Out (LIFO) principle.

Q2. What is Stack Overflow and Stack Underflow

- * Stack Overflow: Attempting to push an element onto a completely filled stack ($\text{top} == \text{MAX} - 1$).

Q3. Define Queue ADT and state FIFO principle.

A Queue is a linear data structure following the First-In, First-Out (FIFO) principle, where elements are inserted at the Rear end and deleted from the Front end.

Q4. What is the advantage of a Circular Queue over a Linear Queue

In a Linear Queue, space freed by dequeue cannot be reused once $\text{rear} == \text{MAX} - 1$ (false overflow). A Circular Queue wraps pointers around using modulo arithmetic $((\text{rear} + 1) \% \text{MAX})$, eliminating memory wastage.

Q5. What is a Double-Ended Queue (Deque)

A Deque is a generalized queue where insertion and deletion operations can be performed efficiently from both ends (Front and Rear).